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EE-102, Section 003

Term Project

Cocktail Machine

PROJECT PROPOSAL

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Cocktail Machine

Name: Furkan Büyüksarıkulak

Course Code: EE 102

Section: 003

OBJECTIVE:

In this project we are planning to create a functional work by using Basys3. This work I aim for is a machine that will mix various drinks and servet hem in different combinations. I will use Basys3 to servet he drinks in line with the request from the user and to establish the operation and connection of the electronic equipment to be used in this process.

PHASES:

This Project will consist of 2 phases, the progress I have planned for the progress demo and final demo phases are as follows:

- 1- In the first phase, I aim to focus on the mechanical formation of my project. Things included in this; To create the internal skeleton from the wood that forms the basis, to cover the skeleton with PVC material, to provide the necessary materials such as peristaltic pump, pipe, digital screen, and to learn and control the operation of the supplied materials.
- 2- In the final demo phase, I will focus more on the electronic and software side of the machine. Assembly and functionalization of the parts we have are also included in this process. When this phase is finished, I aim to obtain a functional machine that fills the selected mixtures into a glass.

EQUIPMENTS:

- Wooden frame
- PVC sheet
- DC 12V peristaltic pump
- LCD display
- Push buttons
- DC socket module
- Silicone pipe
- Sensor
- Basys3
- Cup

DETAILED DESCRIPTION OF PROJECT:

This project aims to make a machine that can serve mixed drinks by using VHDL programming language on Basys3 FPGA board. I plan to serve mixes of 3 different drinks in different combinations. These combinations will involve filling a full glass with one, two or three types of beverages. It will be up to the user to choose which drink mixes to fill by using push buttons and LCD display. I will start the project by creating the frame of the machine by using wooden pieces. Afterwards, I will cut PVC sheet according to the dimensions of the skeleton I created. I will assemble them after determining the places where the motors, sensors and pipes will enter on the PVC and shaping them accordingly. Then, I will examine the operating configurations of the DC peristaltic pump to help me to code Basys3. Namely, I will measure the required voltage and information about how many milliliter of liquid it transfers per second. Then I will move on to the coding phase. By trial and error, I will discover how many seconds it takes to fill half of the glass and depending on this, I will keep the peristaltic pumps of selected one active during that seconds depending on the selections. Using this information, the pumps will work sequentially, depending on which drinks are selected on the screen I will assemble. There will also be a sensor that will receive information about whether the glass is at the filling station or not, which will prevent filling without a glass. When all these are combined, beverage tanks can be anywhere in the invisible part of the system because this can be achieved by extending the pipes coming from the peristaltic pumps and this provides a simpler appearance. Finally, for example, when you choose a mixture of orange and lemon juice, first half of the glass will be filled with orange juice, then the other half will be filled with lemon juice, and you will receive feedback on the screen that it is full and you can take it. After the drink is filled, options such as napkins, extra sugar and spices will be accessible manually from the station next to the machine. The draft 3D models (Figure 1) of the machine that is the subject of the project are as follows.

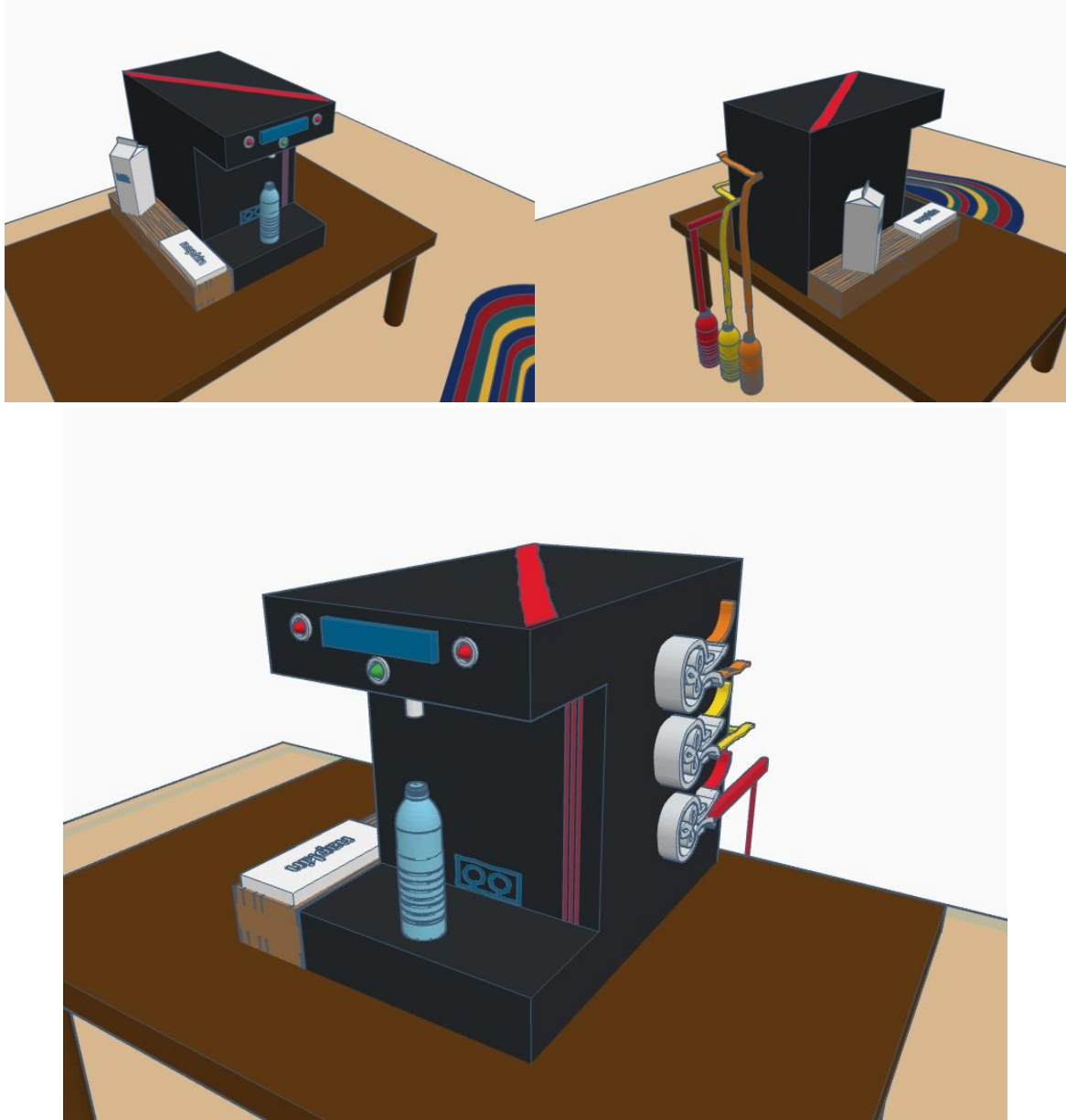


Figure 1 : 3D model of cocktail machine